

For the following formulas, assume

- * $\mathbf{F}$ a vector-valued function with continuous partial derivatives.
- * g a scalar-valued function with continuous partial derivatives.
- * $\mathbf{r}(t)$ represents some parameterization of a curve C
- * $\mathbf{r}(u, v)$ represents some parameterization of a surface S

Operators:

$$* \nabla g(x, y, z) = [g_x, g_y, g_z] \quad * \operatorname{div} \mathbf{F} = M_x + N_y + P_z \quad * \operatorname{curl} \mathbf{F} = [P_y - N_z, M_z - P_x, N_x - M_y]$$

Stokes' Theorem: Let C be a closed curve which is the boundary for a surface S . Then

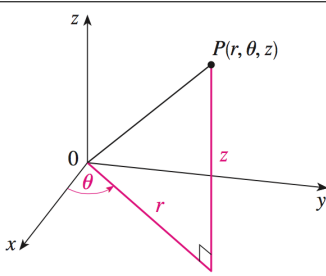
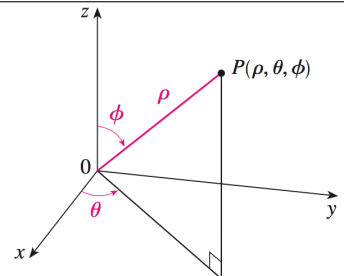
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dA$$

* Recall: Green's Theorem is a special case of Stokes' Theorem where S is in the xy -plane and $\mathbf{n} = [0, 0, 1]$.

Divergence Theorem: Let S be a surface that is the boundary of T . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_T \operatorname{div} \mathbf{F} dV$$

Special Coordinates

Cylindrical		Spherical	
$x = r \cos \theta$		$x = \rho \sin \phi \cos \theta$	
$y = r \sin \theta$		$y = \rho \sin \phi \sin \theta$	
$z = z$		$z = \rho \cos \phi$	
$x^2 + y^2 = r^2$		$x^2 + y^2 + z^2 = \rho^2$	
$dV = r dr d\theta dz$		$dV = \rho^2 \sin \phi d\rho d\theta d\phi$	

Complex Numbers:

$$\text{Let } z = re^{\theta i} = r(\cos \theta + i \sin \theta)$$

$$z^n = r^n e^{n\theta i} = r^n [\cos(n\theta) + i \sin(n\theta)]$$

$$z^{1/n} = r^{1/n} e^{\frac{\theta + 2\pi k}{n} i} = r^{1/n} \left[\cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right] \quad \text{for } k = 0, 1, 2, \dots, n-1$$

Fourier Series:

The Fourier series of a function, $f(x)$, with period, $2L$, is given by

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \left(\frac{n\pi x}{L} \right) + b_n \sin \left(\frac{n\pi x}{L} \right) \right]$$

where

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx, \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \left(\frac{n\pi x}{L} \right) dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \left(\frac{n\pi x}{L} \right) dx$$